
X-AVDD: Cross-Attentive Audio-Visual Dataset Distillation

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Abstract

Audio-visual dataset distillation aims to synthesize a small dataset from a large paired audio-visual dataset, while nearly preserving the training utility for neural networks. Existing approaches such as DM and AVDD face two critical limitations: 1) they are memory-intensive, making them difficult to scale to complex architectures and higher samples-per-class (SPC) settings, and 2) they often fail to preserve strong cross-modal semantic correspondence. To tackle these issues, we propose **X-AVDD**, a **cross-attention-based** distillation method that explicitly captures inter-modal relationships during synthesis. By decoupling cross-modal alignment from joint optimization, X-AVDD is substantially more memory-efficient, reducing end-to-end synthesis cost by $\sim 8\times$ GPU-hours, while producing highly aligned synthetic data. Empirically, X-AVDD significantly outperforms other audio-visual distillation methods. On VGGSound, an audio-visual dataset of short YouTube clips, X-AVDD improves top-1 test accuracy from 8.2% $\rightarrow$ 24.0% at SPC=10 and 9.8% $\rightarrow$ 26.5% at SPC=20. Furthermore, X-AVDD improves cross-architecture generalization (e.g., 25.5% $\rightarrow$ 35.3% for a ViT target on AVE dataset at SPC = 50) and successfully scales to complex architectures such as AudioCLIP, achieving 62.3% accuracy using only 5.65% of the full dataset, against the full dataset accuracy of 72.4% while completing in $9\times$ lower training time compared to the full dataset.

1 Introduction

Modern multimodal models derive much of their capability from training on massive datasets that jointly capture complementary signals across modalities (e.g., vision, audio, text) [Qin et al., 2025]. While such datasets enable richer representations and improved robustness over single-modality learning, their scale creates a growing bottleneck: storing, curating, and repeatedly training on multimodal corpora is expensive in both computation and energy [Schwartz et al., 2020, Strubell et al., 2019]. Recent work has therefore increasingly emphasized *data-centric* methods to improve data quality, and data curation as a key lever for efficiently scaling multimodal models [Qin et al., 2025].

Dataset distillation (DD), also known as Dataset Condensation, is an increasingly popular route to data-efficient learning. DD methods synthesize a

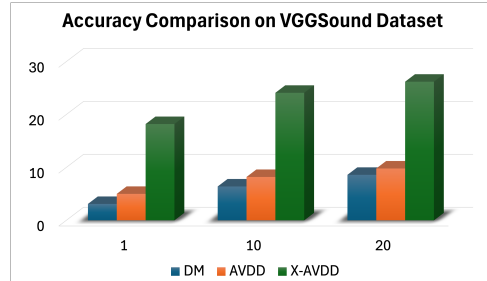


Figure 1: Accuracy on the VGGSound dataset under different samples-per-class settings. X-AVDD consistently outperforms DM and AVDD [Kushwaha et al., 2024].

small set of labeled synthetic examples such that models trained on the synthetic data achieve performance comparable to those trained on the full dataset [Wang et al., 2018, Zhao et al., 2021, Sachdeva and McAuley, 2023, Liu and Du, 2025]. Distilled datasets enable rapid iteration for tasks like neural architecture search and hyperparameter tuning [Ding et al., 2024, Such et al., 2020], and provide a compact interface for sharing training signals while maintaining privacy [Dong et al., 2022].

As DD moves beyond unimodal settings (primarily image classification), *multimodal* distillation [Wu et al., 2024] introduces a central additional challenge: the synthetic data must preserve not only modality-specific information, but also the cross-modal associations that make multimodal learning effective. In this paper, we focus on the multimodal distillation problem of *audio-visual dataset distillation* introduced in [Kushwaha et al., 2024]. Given a large-scale paired audio-visual dataset, we want to synthesize a much smaller dataset such that downstream models trained on the synthetic dataset have good performance. Following [Kushwaha et al., 2024], we consider audio-visual event recognition, which is the closest audio-visual analogue of image classification, as the main proxy task for model evaluation. Their method AVDD, extends Distribution Matching (DM) [Zhao and Bilen, 2023] to paired audio-visual data. However, their approach faces two fundamental limitations:

Memory Inefficiency. It is highly memory-intensive, making it difficult to scale distillation to higher samples-per-class (SPC) settings and complex modern architectures.

Modality Gap. It fails to preserve strong semantic correspondence between modalities, leading to a high “modality gap” [Liang et al., 2022, Zhang et al., 2025]. As shown in Figure 2, AVDD generates synthetic data with significantly higher class-wise ℓ_2 distances between audio and visual features compared to the original dataset.

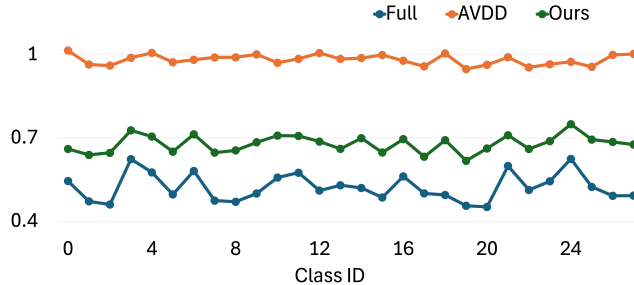


Figure 2: Modality gap (class-wise ℓ_2 distance between audio and visual class centers in the feature space) on the AVE dataset, and datasets distilled by AVDD and X-AVDD using AudioCLIP network. X-AVDD achieves a consistently lower modality gap.

To address these issues, we propose *X-AVDD*, a *cross-attentive* audio-visual dataset distillation method designed for scalability and stronger cross-modal coupling. The key idea is to *explicitly capture inter-modality relationships during distillation* using bi-directional, multi-head cross-attention, while *decoupling cross-modal alignment from joint optimization* via a pre-trained teacher model.

This design substantially improves efficiency: X-AVDD reduces total synthesis time on a Tesla V100 GPU from **10 hours** to **1.3 hours**. Figure 1 presents the impressive generalization performance obtained by X-AVDD compared to the baseline DM and AVDD methods.

Contributions.

- We introduce **X-AVDD**, a cross-attention-based method that explicitly models inter-modal relationships during audio-visual dataset distillation.
- We propose a **decoupled cross-modal alignment strategy** using bi-directional cross-attention, yielding substantially improved **memory efficiency** and enabling scaling to larger SPC and modern architectures.
- We demonstrate that X-AVDD produces **better-aligned synthetic data** (lower modality gap) and delivers consistent improvements over AVDD and multimodal SRe²L baselines across multiple benchmarks.

2 Related Works

2.1 Dataset Distillation

Dataset Distillation (DD), also referred to as Dataset Condensation, aims to synthesize a compact set of labeled examples that preserves the training signal of a much larger original dataset. While early methods rely on bi-level optimization [Wang et al., 2018] (e.g., gradient matching [Zhao et al., 2021, Zhao and Bilen, 2021], trajectory matching [Cazenavette et al., 2022, Cui et al., 2023]), which are quite memory and compute intensive. To improve scalability, distribution-matching methods [Zhao and Bilen, 2023] align the statistics of real and synthetic embeddings. However, these methods often encounter memory bottlenecks as the synthetic budget or model complexity increases. We refer the reader to recent surveys for broader coverage and taxonomies of DD methods Sachdeva and McAuley [2023], Lei and Tao [2023], Liu and Du [2025].

2.2 Attention in DD

While attention has been explored in unimodal distillation, such as DataDAM [Sajedi et al., 2023], which matches spatial attention maps, our work is the first to employ bi-directional cross-attention in a multimodal context. Unlike DataDAM’s focus on unimodal spatial patterns, X-AVDD uses bi-directional multi-head cross-attention to explicitly model the temporal and semantic dependencies between audio and visual streams during synthesis.

2.3 Multimodal Dataset Distillation (MDD)

MDD extends distillation to settings requiring the preservation of cross-modal associations. While vision-language distillation [Wu et al., 2024, Xu et al., 2024, Zhang et al., 2025, Zhao et al., 2025, Lee and Chung, 2025] has seen rapid progress, audio-visual distillation remains under-explored. A primary challenge in MDD is the “modality collapse”, in which synthetic pairs lose semantic alignment [Zhang et al., 2025]. Our work is the first to address this in the audio-visual domain by using a decoupled alignment strategy to proactively minimize this gap.

2.4 Audio-Visual Event Recognition

Recognizing events in unconstrained videos requires learning aligned representations across audio and visual modalities [Brousseau et al., 2022]. Benchmark datasets such as AVE [Tian et al., 2018] and VGGSound [Chen et al., 2020] have driven the development of complex architectures, such as AudioCLIP [Guzhov et al., 2022]. Contrastive learning and cross-modal attention have become prominent directions for learning aligned representations across modalities [Arandjelovic and Zisserman, 2017, Miech et al., 2019]. However, the computational cost of training these models on the full dataset is quite high. Our work contributes to this domain by making audio-visual training more data-efficient through distillation, enabling faster iteration and cheaper training in this setting.

2.5 Limitations of AVDD

Despite establishing the first framework for audio-visual dataset distillation, AVDD faces a fundamental scalability bottleneck due to its reliance on a non-decoupled Distribution Matching (DM) paradigm. In AVDD, the entire set of synthetic samples must be held in GPU VRAM simultaneously during the optimization loop to match the global feature distributions, leading to Out-of-Memory (OOM) failures as the SPC budget or model complexity increases. While a class-wise batching strategy could theoretically reduce memory, our empirical tests show that it drastically degrades performance (e.g., a drop from 58.0% to 49.5% at SPC=20), because it loses the inter-class relationship information necessary for effective distribution matching. Consequently, AVDD is largely restricted to simple ConvNet architectures and lower SPC settings, failing to generalize to modern high-parameter models like ViT [Dosovitskiy et al., 2021] or AudioCLIP [Guzhov et al., 2022]. Furthermore, AVDD involves a “buffer creation” cost (reflected in our AVDD timing breakdown in Table 5).

X-AVDD addresses these limitations by decoupling the alignment phase from the synthesis loop, enabling class-wise optimization via frozen, cross-attentive-guided networks that maintain superior accuracy while reducing peak VRAM usage and total synthesis time.

3 Methodology

In this section, we describe our proposed X-AVDD in detail. Table 10 (in Appendix A) provides the notation used throughout the paper.

Problem formulation. Let the labeled dataset be $\mathcal{V} = \{(x_i^a, x_i^v, y_i)\}_{i=1}^N$, where $x_i^a \in \mathcal{X}^a$ and $x_i^v \in \mathcal{X}^v$ denote the audio and visual modalities of the i -th sample, respectively, and $y_i \in \mathcal{Y}$ is the corresponding ground-truth label. The visual input x_i^v is represented by the central frame of the video, while the audio input x_i^a is represented by the corresponding spectrogram.

The goal of DD is to learn a compact synthetic dataset $\mathcal{S} = \{(\tilde{x}_i^a, \tilde{x}_i^v, \tilde{y}_i)\}_{i=1}^M$, $M \ll N$, such that a model parameterized by $\theta^{\mathcal{S}}$, trained on $\mathcal{S}$, achieves generalization performance comparable to a model $\theta^{\mathcal{V}}$ trained on the full dataset $\mathcal{V}$.

Cross Attention. This is a fundamental attention mechanism, which allows different modalities to exchange information in a multi-modal setting [Tsai et al., 2019]. Information exchange occurs through queries from one modality attending to keys and values from another modality. Cross-attention enables fine-grained semantic alignment across different input streams. Intuitively, for audio-visual dataset distillation, cross-attention can capture complementary information [Praveen et al., 2022, Zeng et al., 2023] between the two modalities and guide distillation steps towards generating synthetic samples that capture informative semantic information from the real dataset.

Some recent works [Hiller et al., 2024, Si et al., 2025] have explored bidirectional cross-attention for training models on multimodal datasets. Bi-directional cross-attention is a mechanism in multimodal models in which two modalities (e.g., text and images, or audio and video) attend to each other in both directions, allowing each to refine its representation based on the other. Concretely, one attention pass lets modality A query and extract relevant information from modality B, while a second, symmetric pass lets modality B query modality A. This mutual exchange captures richer, more aligned relationships than one-way fusion. This bidirectional interaction improves tasks such as alignment, retrieval, and generation by modeling inter-dependencies rather than treating one modality as dominant, leading to more coherent, semantically grounded multimodal representations. Figure 3 shows the block diagram of the implementation of the bi-directional cross-attention mechanism.

Formulation of X-AVDD. In this paper, we propose *X-AVDD*, a cross-attention-based decoupled dataset distillation method. Our proposed method decouples distillation of synthetic data from training of teacher model with original dataset. This decoupling ensures distillation process can be parallelized and does not depend upon having access to the original large dataset. We incorporate a bidirectional cross-attention module that enables the extraction of complementary information from both modalities. Our method consists of three steps.

1. Pre-training of the teacher network, which incorporates bi-directional cross-attention.
2. Computation (and storage) of real embeddings for both audio and visual modalities through the cross-attention layers.
3. Iterative learning of a synthetic dataset by matching cross-attended embeddings from real and synthetic samples.

In our approach, we use randomly sampled audio and frame pairs from real data to initialize the distillation process (similar to Cazenavette et al. [2022] and Du et al. [2024]). For the experiments, the same set of initialization samples is used for both the baseline and our proposed method. We use a Euclidean Distribution Matching (EDM) loss function that encourages synthetic embeddings to cluster near real ones in Euclidean space by minimizing the pairwise Euclidean distances between attended embeddings from real and synthetic samples. Pairwise Euclidean distance between two matrices $\mathbf{A}$ and $\mathbf{B}$ of dimensions (M, d) and (N, d) respectively, is calculated as:

$$D_{\text{euc}}(\mathbf{A}, \mathbf{B}) = \sum_{j=1}^N \sum_{i=1}^M \|\mathbf{A}[i, :] - \mathbf{B}[j, :]\|_2 \quad (1)$$

Let $\{x_{ar}^C, x_{vr}^C\}$ and $\{x_{as}^C, x_{vs}^C\}$ be the real and synthetic audio-image sample pairs belonging to class C , respectively. Let h_a and h_v be audio and visual encoders, and τ_a and τ_v be the tokenization operators for audio and visual modality, respectively. The spatial tokens for real and synthetic samples

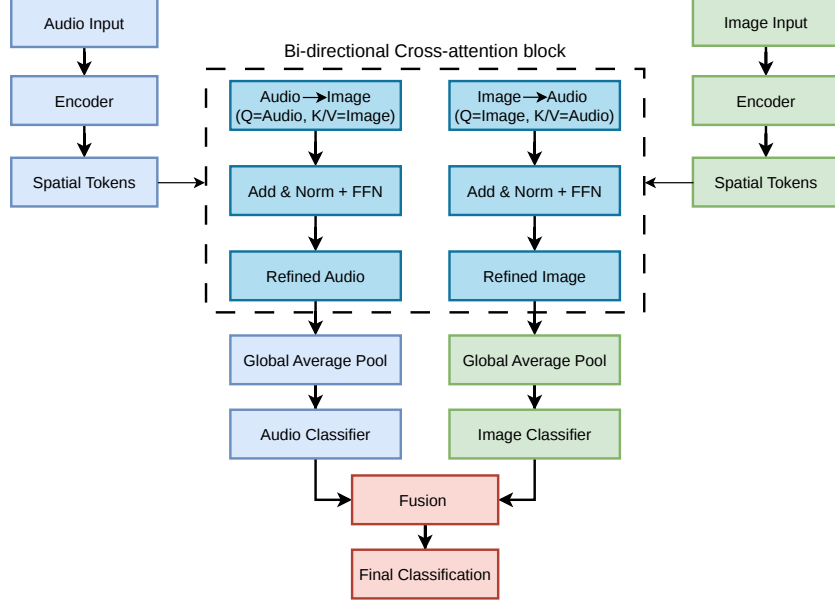


Figure 3: Overview of the bi-directional cross-attention mechanism for Audio-Visual event recognition.

are expressed as:

$$\mathbf{A}_r^C = \tau_a(h_a(x_{ar}^C)) \quad \mathbf{V}_r^C = \tau_v(h_v(x_{vr}^C)) \quad (2)$$

$$\mathbf{A}_s^C = \tau_a(h_a(x_{as}^C)) \quad \mathbf{V}_s^C = \tau_v(h_v(x_{vs}^C)) \quad (3)$$

Let $\mathcal{T}_a$ and $\mathcal{T}_v$ be audio $\rightarrow$ image and image $\rightarrow$ audio branches of the bi-directional cross-attention block, respectively. The output of the Global Average Pool (GAP) is the attended embeddings of corresponding real data, which are stored and utilized during the distillation process. Let g_a and g_v be the GAP operators for the audio and visual branches, respectively. The attended embeddings for real data can be represented as :

$$\mathbf{E}_{r(A \rightarrow V)}^C = g_a(\mathcal{T}_a(\mathbf{A}_r^C, \mathbf{V}_r^C)) \quad \mathbf{E}_{r(V \rightarrow A)}^C = g_v(\mathcal{T}_v(\mathbf{A}_r^C, \mathbf{V}_r^C)) \quad (4)$$

The attended embeddings for the synthetic data are given by:

$$\mathbf{E}_{s(V \rightarrow A)}^C = g_v(\mathcal{T}_v(\mathbf{A}_s^C, \mathbf{V}_r^C)) \quad \mathbf{E}_{s(A \rightarrow V)}^C = g_a(\mathcal{T}_a(\mathbf{A}_r^C, \mathbf{V}_s^C)) \quad (5)$$

Note that the audio-guided image embeddings $\mathbf{E}_{s(A \rightarrow V)}^C$ are computed by using synthetic image as query and real audio as key and value. Similarly, the image-guided audio embeddings $\mathbf{E}_{s(V \rightarrow A)}^C$ are computed by using synthetic audio as query and real image as key and value. As stated earlier, in order to encourage synthetic embeddings to cluster near real ones, we perform Euclidean distribution matching (EDM), defined as:

$$l_{A \rightarrow V}^C = D_{\text{euc}}(\mathbf{E}_{r(A \rightarrow V)}^C, \mathbf{E}_{s(A \rightarrow V)}^C) \quad (6)$$

$$l_{V \rightarrow A}^C = D_{\text{euc}}(\mathbf{E}_{r(V \rightarrow A)}^C, \mathbf{E}_{s(V \rightarrow A)}^C) \quad (7)$$

The complete loss function during the synthesis stage is:

$$l^C = \lambda_{ce} l_{ce}^C + \lambda_{av} l_{(A \rightarrow V)}^C + \lambda_{va} l_{(V \rightarrow A)}^C \quad (8)$$

Where l_{ce}^C is the standard cross-entropy loss of the classification task, and λ_{ce} , λ_{av} , and λ_{va} are corresponding weights of the loss functions.

4 Experiments and Results

Datasets and protocol. We evaluate our method on four benchmark audio-visual datasets: AVE [Tian et al., 2018], MUSIC-21 [Zhao et al., 2019], VGGSound [Chen et al., 2020], and VGGSound-10K (10-class subset released with AVDD [Kushwaha et al., 2024]). We use AVDD’s preprocessing and splits for fair comparison, and report mean and standard deviation of test accuracy over five seeds (0–4). See Appendix E for dataset and preprocessing details.

Backbone architecture. Following AVDD, we first employ a ConvNet architecture [Zhao et al., 2021] for both audio and video encoders. Additionally, we employed more complex architectures, such as ResNet-18 [He et al., 2016], ViT [Dosovitskiy et al., 2021], and AudioCLIP [Guzhov et al., 2022], to thoroughly evaluate X-AVDD’s applicability.

Cross-architecture comparison To validate the effectiveness of the synthetic dataset, we train randomly initialized models, which are different from the backbone architecture utilized to generate the synthetic images. We used ResNet-18 to generate synthetic images, and VGG-11 [Simonyan and Zisserman, 2015] and ViT [Dosovitskiy et al., 2021] models for cross-architecture comparison.

Baselines. For the ConvNet architecture, we compared AVDD-reported results with those obtained by our proposed method. For the other architectures, AVDD results in an OOM issue due to hardware limitations. We instead designed and implemented a new multimodal adaptation of SRe²L [Yin et al., 2023], as a baseline for distillation, and compared it against X-AVDD. Details of the multimodal adaptation of SRe²L are provided in Appendix D. Hyperparameter details are provided in Appendix F.

4.1 Results for ConvNet, ResNet-18 and ViT architectures

Table 1 provides test set accuracies obtained by training ConvNet, ResNet-18, and ViT architectures from scratch using the distilled dataset for various Samples Per Class (SPC) settings. We can see that X-AVDD significantly outperforms AVDD across most SPC settings. In the ConvNet architecture, at extremely low SPC, the extra cross-modal constraints can over-regularize, whereas benefits emerge as soon as synthetic diversity increases ($\text{SPC} \geq 10$). There is also a particularly significant jump across all SPCs for VGGSound, which is by far the largest dataset used in our experiments.

Table 1: Comparison of accuracy (%) on synthetic datasets for various architectures with the ensemble fusion method. For the ConvNet architecture, comparisons are carried out against DM and AVDD, while for ResNet-18 and ViT, comparisons are carried out against a multimodal implementation of SRe²L. SPC denotes samples per class. Bold indicates best performance within each architecture group. - indicates Out-of-Memory (OOM) errors.

Dataset	SPC	Ratio (%)	ConvNet			ResNet-18		ViT	
			DM	AVDD	X-AVDD	SRe ² L	X-AVDD	SRe ² L	X-AVDD
VGGS-10K	1	0.11	36.5 ± 2.5	40.4 ± 1.8	37.6 ± 0.8	21.7 ± 0.6	30.0 ± 1.5	16.7 ± 0.9	20.2 ± 0.5
	10	1.13	43.8 ± 1.7	55.0 ± 1.7	57.2 ± 0.9	46.9 ± 2.0	54.5 ± 0.7	25.4 ± 0.5	30.0 ± 0.7
	20	2.26	49.0 ± 2.4	58.0 ± 1.7	62.7 ± 0.5	52.7 ± 0.5	61.5 ± 0.8	26.1 ± 1.8	35.9 ± 0.8
	50	5.65	-	-	66.1 ± 0.3	60.4 ± 0.9	66.7 ± 0.9	30.2 ± 1.2	42.1 ± 1.2
	Full			74.7		77.4		62.1	
AVE	1	0.1	21.7 ± 1.5	23.0 ± 1.4	22.0 ± 0.4	18.5 ± 1.0	20.3 ± 0.8	11.2 ± 0.8	12.4 ± 0.7
	10	1.0	28.1 ± 1.8	36.8 ± 0.9	43.5 ± 0.2	41.2 ± 0.5	46.9 ± 0.7	24.5 ± 0.7	26.7 ± 0.1
	20	2.0	32.6 ± 1.0	40.1 ± 1.0	50.2 ± 0.8	45.9 ± 0.3	50.3 ± 0.3	25.3 ± 2.3	31.1 ± 0.8
	50	5.0	-	-	56.2 ± 0.6	49.1 ± 0.3	56.3 ± 0.6	27.8 ± 1.3	33.0 ± 0.4
	Full			58.5		59.2		47.9	
MUSIC-21	1	0.014	38.3 ± 1.3	44.0 ± 2.2	47.9 ± 0.7	21.6 ± 0.8	30.8 ± 2.1	13.6 ± 1.3	21.0 ± 0.5
	10	0.14	54.8 ± 1.4	68.1 ± 1.0	74.5 ± 0.5	58.7 ± 1.4	70.9 ± 0.3	37.4 ± 0.3	44.5 ± 0.1
	20	0.28	61.1 ± 1.3	70.3 ± 0.7	77.6 ± 0.9	61.2 ± 2.9	74.1 ± 0.4	41.6 ± 0.9	48.0 ± 1.4
	50	0.70	-	-	79.7 ± 0.3	71.9 ± 0.6	78.4 ± 0.3	52.2 ± 1.2	56.7 ± 0.8
	Full			80.5		81.4		71.3	
VGGSound	1	0.18	3.1 ± 0.2	5.0 ± 0.3	18.3 ± 0.3	7.9 ± 0.9	10.9 ± 0.2		
	10	1.87	6.4 ± 0.3	8.2 ± 0.2	24.0 ± 0.1	17.3 ± 1.4	22.4 ± 0.5		
	20	3.74	8.6 ± 0.1	9.8 ± 0.3	26.5 ± 1.1	21.1 ± 0.3	25.1 ± 0.1		
	50	9.35	-	-	28.2 ± 0.7	22.2 ± 0.5	28.4 ± 0.9		
	Full			30.1		32.1			

4.2 Results for AudioCLIP architecture

AudioCLIP [Guzhov et al., 2022] extends CLIP to the Audio modality along with image and text, incorporating ESResNeXt audio-model into the CLIP framework. Table 2 provides a comparative analysis of accuracy values obtained on the test set of VGGSound-10K and AVE datasets with synthetic images generated by our proposed method. With a fraction of the time the model takes to train on the full dataset, distilled images can achieve comparable generalization accuracy, underscoring the role of distillation in training an audio-visual dataset with SOTA architectures.

Table 2: Test accuracy obtained by synthetic data generated by X-AVDD on VGGSound-10K and AVE datasets with AudioCLIP architecture. Ratio is the percentage of the condensed image as compared to the full dataset. With a fraction of the time taken to train the model on the full dataset, distilled images can achieve comparable generalization accuracy, underscoring the role of distillation for training audio-visual datasets with SOTA architectures.

SPC	VGGSound-10K			AVE		
	Ratio (%)	Accuracy (%)	Train time (Hrs.)	Ratio (%)	Accuracy (%)	Train time (Hrs.)
1	0.11	24.8 $\pm$ 0.6	0.28	0.1	44.0 $\pm$ 1.2	0.43
10	1.13	33.7 $\pm$ 0.3	0.56	1.0	74.5 $\pm$ 0.9	1.31
20	2.26	46.8 $\pm$ 0.9	1.16	2.0	76.8 $\pm$ 1.1	2.66
50	5.65	62.3 $\pm$ 0.8	2.74	5.0	81.3 $\pm$ 0.9	6.43
Full	100	72.4 $\pm$ 0.7	25.30	100	85.6 $\pm$ 1.1	72.12

4.3 Cross architecture analysis

To study the generalization capability of the generated synthetic dataset, we carried out a cross-architecture study with two different architectures compared to the base model through which the synthetic dataset is generated. We utilized VGG-11 [Simonyan and Zisserman, 2015], and transformer-based architecture, ViT [Dosovitskiy et al., 2021]. The results are listed in Table 3. Our proposed method significantly outperforms the baseline across various datasets and SPC settings. Significant improvement over the baseline on the MUSIC-21 dataset with ViT as the target architecture provides further proof of the efficacy of our proposed method.

Table 3: Comparative analysis of cross-architecture generalization performance of the distilled dataset generated using the ResNet-18 architecture for both audio and video modality. We experimented with two target architectures: VGG11 and ViT. For both, the synthetic dataset generated by X-AVDD achieves higher generalization accuracy than that generated by the SRe²L adaptation. Results are provided for the Ensemble fusion method.

Target	SPC	VGGSound-10K		AVE		Music-21	
		SRe ² L	X-AVDD	SRe ² L	X-AVDD	SRe ² L	X-AVDD
VGG-11	1	14.6 $\pm$ 3.7	20.4 $\pm$ 2.3	6.2 $\pm$ 2.5	11.4 $\pm$ 0.7	18.2 $\pm$ 1.9	22.3 $\pm$ 1.2
	10	34.8 $\pm$ 1.3	41.3 $\pm$ 2.2	21.8 $\pm$ 0.4	29.9 $\pm$ 0.7	39.7 $\pm$ 1.6	47.3 $\pm$ 2.6
	20	44.0 $\pm$ 2.4	51.5 $\pm$ 0.8	29.1 $\pm$ 0.3	35.6 $\pm$ 0.6	42.8 $\pm$ 1.1	58.7 $\pm$ 1.9
	50	53.2 $\pm$ 0.2	59.4 $\pm$ 0.2	35.0 $\pm$ 0.7	42.4 $\pm$ 0.3	48.5 $\pm$ 0.5	64.3 $\pm$ 1.1
ViT	1	13.9 $\pm$ 1.1	23.3 $\pm$ 0.9	12.6 $\pm$ 0.8	13.9 $\pm$ 0.1	10.0 $\pm$ 0.5	14.4 $\pm$ 1.8
	10	30.0 $\pm$ 2.9	39.9 $\pm$ 0.8	21.0 $\pm$ 0.4	27.2 $\pm$ 1.1	26.7 $\pm$ 2.0	44.5 $\pm$ 0.8
	20	38.6 $\pm$ 0.6	46.2 $\pm$ 0.7	23.1 $\pm$ 0.3	30.4 $\pm$ 0.7	28.8 $\pm$ 0.7	45.1 $\pm$ 0.9
	50	44.3 $\pm$ 1.1	51.3 $\pm$ 1.4	25.5 $\pm$ 2.4	35.3 $\pm$ 0.5	29.0 $\pm$ 0.9	56.5 $\pm$ 1.3

4.4 Transfer learning with frozen backbone of SOTA image and audio encoder models

In this experiment, we investigate whether distilled datasets generated using simpler models such as ConvNet and ResNet-18, are sufficient to train lightweight classifiers on top of frozen, state-of-the-art image and audio encoder models. We used two complementary setups for this experiment. The first setup uses the audio and visual backbones of ImageBind [Girdhar et al., 2023], which are designed to map multimodal data onto a shared feature space. The second setup leverages Audio Spectrogram Transformer (AST) [Gong et al., 2021] as the audio encoder and SigLIP2 [Tschannen et al., 2025] as the visual encoder. We chose AST because of its strong performance on music/environmental audio domains on MAEB [Assadi et al., 2026], and chose SigLIP2 [Tschannen et al., 2025] because the

SigLIP family is among the strongest open image embedding models on MIEB [Xiao et al., 2025]. MAEB and MIEB are integrated with the very popular MTEB benchmark [Muennighoff et al., 2023]. Small projection heads and linear classifiers are then trained on the extracted, frozen features using only the distilled synthetic data.

We compare the performance of data distilled by AVDD and X-AVDD using ConvNet, and the data distilled by SRe²L and X-AVDD using ResNet-18. Table 4 tabulates these results for the VGGSound-10K dataset. With only 50 samples per class, the dataset distilled by X-AVDD using ResNet-18 achieves significant performance compared to training with the full dataset, highlighting the capability of X-AVDD and its adaptation to powerful pretrained foundation models for downstream audio-visual classification tasks.

Table 4: Performance analysis of distilled dataset for transfer learning with frozen backbone of SOTA models. X-AVDD (C) and X-AVDD (R) denote the synthetic datasets generated by ConvNet and ResNet-18, respectively. With only 50 samples per class, synthetic data generated by the ResNet-18 architecture and X-AVDD achieves performance comparable to that of training on the full dataset.

SPC	ImageBind				AST + SigLIP2			
	AVDD	X-AVDD (C)	SRe ² L	X-AVDD (R)	AVDD	X-AVDD (C)	SRe ² L	X-AVDD (R)
1	20.1 $\pm$ 0.4	25.7 $\pm$ 0.8	25.2 $\pm$ 0.6	28.3 $\pm$ 0.4	23.1 $\pm$ 1.3	24.3 $\pm$ 0.3	18.1 $\pm$ 1.2	37.3 $\pm$ 0.5
10	26.4 $\pm$ 0.9	40.9 $\pm$ 0.6	40.1 $\pm$ 0.8	58.4 $\pm$ 0.5	36.1 $\pm$ 1.0	42.2 $\pm$ 0.7	49.8 $\pm$ 0.7	58.7 $\pm$ 0.9
20	33.2 $\pm$ 0.5	43.1 $\pm$ 0.4	43.7 $\pm$ 0.3	64.3 $\pm$ 0.5	37.7 $\pm$ 1.3	49.1 $\pm$ 1.6	51.2 $\pm$ 1.2	65.6 $\pm$ 0.6
50	-	48.3 $\pm$ 0.3	56.7 $\pm$ 0.6	67.1 $\pm$ 0.3	-	51.7 $\pm$ 1.1	56.5 $\pm$ 0.7	73.5 $\pm$ 0.3
Full	79.8				81.3			

4.5 Memory comparison

For a fair comparison, we ran both AVDD and X-AVDD on a single 32 GB Tesla V100 GPU for SPC=20 on ConvNet architecture. AVDD must contain all synthetic images during the optimization stage, whereas X-AVDD can generate images batch-wise, drastically reducing memory requirements. Table 5 compares GPU-Hour consumption for generating synthetic datasets by AVDD and X-AVDD. This significant reduction in memory consumption enables X-AVDD to scale to higher SPC settings and larger architectures.

Table 5: GPU-Hour comparison between AVDD and our proposed method for various datasets.

Method	VGGSound-10K	AVE	MUSIC-21	VGGSound
AVDD	165	471	363	5342
X-AVDD	23	51	45	628

4.6 Ablation Studies

4.6.1 Impact of α on test accuracy

α is the weight assigned to the cross-entropy loss from hard labels during the training of the student network with the synthetic dataset, where the total loss is a convex combination of cross-entropy loss from hard labels and KL divergence loss from soft labels. Figure 4 shows that the combination of both hard labels and soft labels results in higher generalization accuracy while training a student model from scratch using the distilled synthetic dataset.

4.6.2 Directionality ablation

We ablate the two cross-modal EDM terms by toggling the weights in Eq. 8. Using either the video $\rightarrow$ audio term ($\lambda_{av} = 0$) or audio $\rightarrow$ video term ($\lambda_{va} = 0$) in a standalone fashion consistently underperforms the full bi-directional objective across SPC regimes. This suggests that bidirectional cross-modal constraints are complementary and important for synthesizing informative paired audio-visual data. Table 6 enumerates these observations on VGGSound-10K dataset.

4.6.3 Improvement Over Random

DD-Ranking [Li et al., 2025] introduced a metric called Improvement Over Random (IOR) to carry out a fair evaluation of the impact of a dataset distillation method. IOR is computed by:

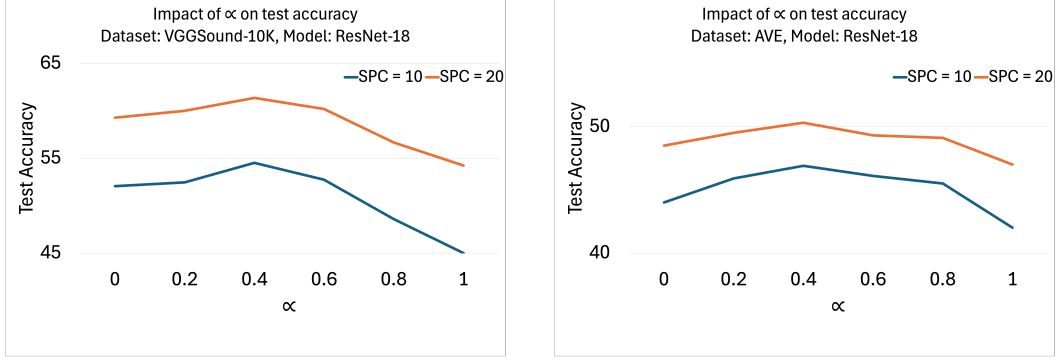


Figure 4: Impact of α on test accuracy. The left panel shows results for the VGGSound-10K dataset and the ResNet-18 architecture, while the right panel shows results for AVE and ResNet-18.

Table 6: Directionality ablation on VGGSound-10K. We toggle the EDM loss weights in Eq. 8.

Method	SPC=1	SPC=10	SPC=20	SPC=50
X-AVDD (both)	30.0 $\pm$ 1.5	54.5 $\pm$ 0.7	61.5 $\pm$ 0.8	66.7 $\pm$ 0.9
Only V $\rightarrow$ A	25.4 $\pm$ 1.3	50.7 $\pm$ 0.1	57.2 $\pm$ 0.6	63.1 $\pm$ 0.1
Only A $\rightarrow$ V	25.9 $\pm$ 1.1	53.4 $\pm$ 0.6	58.5 $\pm$ 1.2	62.0 $\pm$ 0.9

Table 7: IOR values for various SPC settings for different datasets.

Dataset	SPC=1	SPC=10	SPC=20
VGGSound-10K	9.4	6.2	7.2
Music-21	22.3	11.9	9.0
AVE	11.9	14.5	10.1

$IOR = Acc_{syn-any} - Acc_{rdm-any}$, where $Acc_{syn-any}$ and $Acc_{rdm-any}$ are test accuracies of a model trained on the synthetic dataset and a randomly selected subset from the original dataset, respectively. Table 7 enumerates the IOR value for each dataset across each SPC setting. Our proposed method achieves significant improvement over random selection across multiple datasets.

4.6.4 Impact of soft label on generalization accuracy

Table 8 & 9 provides a comparative analysis between the accuracy obtained on training a student network with only hard labels vs. with a combination of hard and soft labels. We can see that training the student network with a combination of hard and soft labels improves test accuracy for both the SRe^2L and our proposed method.

Table 8: Impact of soft label on generalization accuracy (VGGSound-10K, ResNet-18).

SPC	Ratio (%)	SRe^2L (HL)	SRe^2L (SL)	Ours (HL)	Ours (SL)
1	0.11	16.5 $\pm$ 0.1	21.7 $\pm$ 0.6	24.5 $\pm$ 1.5	30.0 $\pm$ 1.5
10	1.13	31.4 $\pm$ 1.3	46.9 $\pm$ 2.0	44.9 $\pm$ 0.2	54.5 $\pm$ 0.7
20	2.26	36.3 $\pm$ 2.1	47.6 $\pm$ 1.6	49.3 $\pm$ 0.9	61.5 $\pm$ 0.8
50	5.65	41.8 $\pm$ 0.7	60.4 $\pm$ 0.9	52.9 $\pm$ 0.3	66.7 $\pm$ 0.9

Table 9: Impact of soft label on generalization accuracy (MUSIC-21, ResNet-18).

SPC	Ratio (%)	SRe^2L (HL)	SRe^2L (SL)	Ours (HL)	Ours (SL)
1	0.014	14.1 $\pm$ 2.7	20.5 $\pm$ 1.0	21.6 $\pm$ 0.8	30.8 $\pm$ 2.1
10	0.14	37.1 $\pm$ 2.8	51.6 $\pm$ 1.1	58.7 $\pm$ 1.4	70.9 $\pm$ 0.3
20	0.28	43.4 $\pm$ 0.5	56.1 $\pm$ 0.1	61.2 $\pm$ 2.9	74.1 $\pm$ 0.4
50	0.70	52.1 $\pm$ 0.6	61.6 $\pm$ 0.4	71.9 $\pm$ 0.6	78.4 $\pm$ 0.3

5 Conclusion

In this work, we introduced X-AVDD, the first scalable audio-visual dataset distillation approach that explicitly uses bi-directional cross-attention between the two modalities to jointly synthesize synthetic audio-visual data. By distilling data through cross-attended embeddings rather than independently matched modality features, X-AVDD preserves semantic alignment between audio and visual streams while substantially reducing memory and computational overhead. Through extensive experiments, we demonstrated the superior performance of X-AVDD across four benchmark audio-visual datasets. We also illustrated the effectiveness of our method by cross-architecture comparison with VGG-11 and ViT. The distilled data generated by X-AVDD shows impressive performance in finetuning frozen backbones of SOTA models, further highlighting the efficacy of our proposed methodology.

Limitations & Future work The present paper focuses on audio-visual classification as the downstream task. Other downstream tasks, such as temporal localization and audio-visual retrieval, would be taken up as future work.

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A Table of notation

Table 10 provides various notation used through the paper.

Table 10: Notation

Symbol	Description
$\mathcal{V}$	Original large-scale training dataset
$\mathcal{S}$	Distilled synthetic dataset
τ_a	Tokenization operator for audio data
τ_v	Tokenization operator for image data
$\mathcal{T}_a$	Audio $\rightarrow$ image branch of bi-directional cross-attention block
$\mathcal{T}_v$	Image $\rightarrow$ audio branch of bi-directional cross-attention block
$\theta^{\mathcal{S}}$	Combined network trained on synthetic dataset
$\theta^{\mathcal{V}}$	Combined network trained on original dataset
η	Learning rate for distillation
l_{ce}	Cross-entropy loss
$\langle x, y \rangle$	Inner product of vector x and vector y
$\ x\ _2$	ℓ_2 -norm (Euclidean-norm) of vector x
τ	Temperature parameter
α	Weight assigned to hard label loss

B Diversity Comparison

Following Mohanty et al. [2026], we use three different metrics for a comprehensive comparative analysis of the diversity of the generated synthetic images:

1. **Vendi Score Friedman and Dieng [2023]**, which is defined as the exponential of the Shannon entropy of the eigenvalues of a similarity matrix. A higher Vendi Score indicates greater diversity in the dataset.
2. **Cosine Similarity**. We compute intra-class pairwise cosine similarity among the embeddings of the synthetic dataset and use the mean as a measure of diversity. For a pair of embeddings $\{x, y\}$, their cosine similarity is defined as:

$$s_{cos}(x, y) = \frac{\langle x, y \rangle}{\|x\| \cdot \|y\|} \quad (9)$$

A lower intra-class cosine similarity indicates higher diversity.

3. **Coverage Naeem et al. [2020]** measures the fraction of real samples whose neighbourhoods contain at least one synthetic sample:

$$\text{coverage} = \frac{1}{N} \sum_{i=1}^N 1_{\{\exists j \text{ s.t. } Y_j \in B(X_i, \text{NND}_k(X_i))\}} \quad (10)$$

where, X is the real dataset, Y is the synthetic dataset, NND_k represents k^{th} Nearest Neighbor Distance, and $B(X, r)$ represents a ball of radius r around the data point X . The distance computations here are carried out in the feature space. A higher coverage indicates greater diversity among the synthetic images.

C Computing Infrastructure Information

Information regarding computing infrastructure for carrying out the experiments is provided in Table 12.

Table 11: Comparative analysis of diversity metrics. Synthetic data generated by X-AVDD performs better across all three diversity metrics as compared to the baseline multimodal adaptation of SRe²L.

Dataset	Method	Vendi $\uparrow$	Similarity $\downarrow$	Coverage $\uparrow$
VGGSound-10k	SRe ² L	1.26	0.89	0.05
	X-AVDD	1.65	0.87	0.15
Music-21	SRe ² L	2.13	0.81	0.03
	X-AVDD	3.04	0.77	0.13
VGGSound	SRe ² L	1.23	0.81	0.05
	X-AVDD	1.71	0.77	0.11

Table 12: Details of the computing infrastructure.

Description	Value
GPU	32 GB Tesla V100
OS	Ubuntu 18.04.1 LTS
Python	3.11.7
Pytorch	2.4.1
NumPy	1.24.3

D Implementation of multimodal SRe²L

Yin et al. [2023] introduced SRe²L, a lightweight tripartite learning paradigm that can scale to larger datasets such as ImageNet-1K. The three stages of SRe²L are:

- **Squeeze**, in which a model is trained on the original training dataset, and the Batch-Norm (BN) layer statistics are stored.
- **Recover**, in which the BN statistics are used to synthesize the condensed dataset.
- **Relabel**, in which the synthetic dataset is assigned soft labels through the model trained on the original dataset.

In this paper, we have adapted this methodology as a baseline for distilling audio-visual datasets. In the Squeeze stage, a teacher network is trained on the full training dataset. The BN statistics stored in various BatchNorm layers in these networks are used during the synthesis stage.

The synthesis stage (Recover) of SRe²L consists of synthesizing the target data starting from independent random Gaussian noise. However, the random nature of Gaussian noise and its lack of semantic information are bottlenecks for effective information distillation. Hence, in our approach, we use real samples for initialization, as do some other dataset distillation methods [Cazenavette et al., 2022, Du et al., 2024]. For a fair comparison, the same initialization samples are used for this baseline and our proposed X-AVDD method.

E Dataset details

We use the same benchmark datasets and the corresponding preprocessing/splits from AVDD Kushwaha et al. [2024].

- **AVE** [Tian et al., 2018]: The Audio-Visual Event (AVE) dataset contains 4,143 unconstrained 10-second YouTube videos spanning 28 event categories with temporally annotated audio-visual event boundaries at second-level granularity (each video contains at least one 2-second audio-visual event). Following AVDD, we segment each clip into non-overlapping 1-second audio-visual windows aligned with the annotations, resulting in train/val/test splits of 27,726 / 3,288 / 3,305 samples.
- **MUSIC-21** [Zhao et al., 2019]: MUSIC-21 is an expanded musical performance dataset containing 1,365 untrimmed videos of musical solos and duets; the original paper reports

a split of 1,065 training videos and 300 test videos, with preprocessing into video shots (5,861 shots in total). Following AVDD, we use their processed split for the *solo* subset, with train/val/test splits of 146,908 / 7,103 / 42,440 samples.

- **VGGSound** [Chen et al., 2020]: VGGSound is a large-scale audio-visual dataset of short YouTube clips curated to ensure audio-visual correspondence. We use the preprocessed train/test split provided by AVDD [Kushwaha et al., 2024], consisting of approximately 165k / 13k train/test samples across 309 classes.
- **VGGSound-10K**: We additionally use the 10-class subset of VGGSound released with AVDD [Kushwaha et al., 2024], which contains 8,808 / 444 train/test samples (using AVDD’s preprocessing and split).

F Hyperparameter details

Table 13 provides the hyperparameters used during pre-training of the teacher network, audio-guided video network, and video-guided audio network. Table 14 tabulates the hyperparameters used during synthesis.

Table 13: Hyperparameter details for pre-training.

Hyperparameter	Value
Epochs	100
Optimizer	Adam
Base learning rate	0.001
Learning rate schedule	cosine decay
Augmentation	RandomResizedCrop
Batch size	128

Table 14: Hyperparameter details for synthesis.

Hyperparameter	Value
Iteration	1000
Optimizer	Adam
Optimizer momentum	$\beta_1, \beta_2 = 0.5, 0.9$
Base learning rate	0.25
Learning rate schedule	cosine decay
Augmentation	RandomResizedCrop
Attn. multiplier	1.0

G Timing comparison

For a fair comparison, we ran both AVDD and our proposed X-AVDD on a single 32 GB Tesla V100 GPU. It may be noted that the training of the teacher network is a one-time process. The synthesis stage of our X-AVDD takes a minuscule amount of time as compared to AVDD.

Table 15: Timing comparison of AVDD and our X-AVDD on VGGSound-10k dataset for synthesis of SPC=20. Kindly note that, while the synthesis stage in AVDD is a recurring process, in our proposed method, training the teacher network and storage of the embeddings are one-time requirements, and only the synthesis stage is the recurring step for generating synthetic samples across various SPC settings.

AVDD		X-AVDD	
Steps	Time	Steps	Time
Buffer creation	1 hr	teacher training & embedding storage	65 mins.
synthesis	9 hrs.	synthesis	12 mins.
Total	10 hrs.	Total	1.3 hrs.

H Modality gap on VGGSound-10K

Figure 5 shows the significant increase in modality gap by synthetic data generated through AVDD on VGGSound-10K dataset. Our proposed method is able to achieve consistently lower modality gap across all classes.

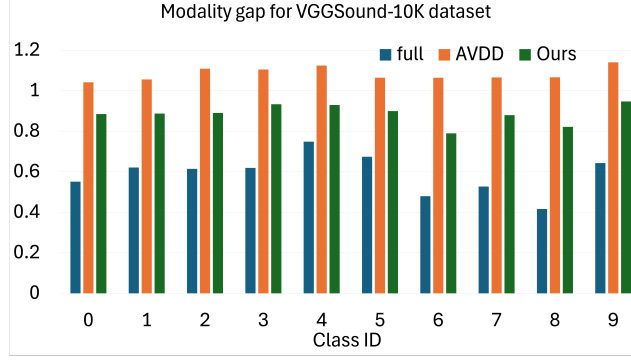


Figure 5: Comparison of modality gap observed in the real dataset, synthetic data generated by AVDD, and our proposed method. As shown, our proposed method achieves a lower modality gap consistently across all classes in the VGGSound-10K dataset.

I Impact of temperature on soft label generation

The temperature parameter τ is vital in smoothening the soft labels Müller et al. [2019]. The impact of τ on test accuracy for the VGGSound-10K dataset with the ResNet-18 architecture for SPC=10 and SPC=20 is depicted in Figure 6.

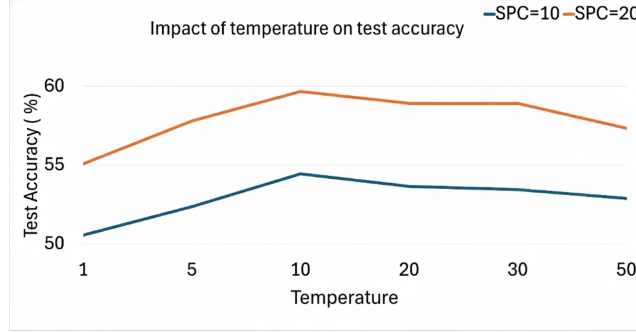


Figure 6: Impact of temperature on test accuracy. Results are provided for VGGSound-10K with ResNet-18 for SPC=10 and SPC=20. We can see that accuracy increases with temperature till it reaches a maximum and then decreases, signifying the importance of the temperature parameter during soft label generation.

J Results with ‘concat’ fusion method

The main paper contains results with the ‘ensemble’ approach of audio-visual fusion. Table 16 compares results obtained by the ‘concat’ fusion method with a synthetic dataset generated by the SRe²L baseline and X-AVDD methods. X-AVDD consistently outperforms the baseline across all four datasets and various SPC settings.

Table 16: Test set accuracy comparison for various datasets with ‘concat’ fusion method.

Dataset	SPC=1	SRe ² L	X-AVDD
VGGSound-10K	1	20.8 $\pm$ 1.5	26.8 $\pm$ 0.7
	10	47.2 $\pm$ 0.7	55.4 $\pm$ 0.7
	20	49.6 $\pm$ 0.2	65.1 $\pm$ 0.2
	50	64.4 $\pm$ 1.2	69.5 $\pm$ 0.3
AVE	1	18.0 $\pm$ 0.1	21.2 $\pm$ 0.4
	10	41.5 $\pm$ 0.6	51.6 $\pm$ 0.4
	20	46.0 $\pm$ 0.7	56.6 $\pm$ 0.2
	50	49.2 $\pm$ 0.4	61.3 $\pm$ 0.2
Music-21	1	29.8 $\pm$ 1.1	34.0 $\pm$ 1.6
	10	66.7 $\pm$ 2.4	70.8 $\pm$ 1.2
	20	67.1 $\pm$ 0.5	73.5 $\pm$ 0.7
	50	72.1 $\pm$ 0.9	79.0 $\pm$ 0.3
VGGSound	1	8.1 $\pm$ 1.2	11.4 $\pm$ 0.9
	10	17.9 $\pm$ 0.8	24.9 $\pm$ 0.3
	20	22.3 $\pm$ 0.1	28.3 $\pm$ 0.2
	50	22.4 $\pm$ 0.6	31.1 $\pm$ 0.1

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